BEAI

A plain-language summary

This note says what each part of the software lets a person do. Then it gives the result that was written down. Then it gives the limit.

The source notes are dated 11–13 October 2026. This summary does not add a measurement. If a value was not written down, the note says it was not recorded.

> This is research software. It studies molecules on a computer.
> It is not medical advice.
> It is not a diagnosis.
> It is not a dosing tool.

What this is

BEAI is research software that runs on one computer. A protein is a large molecule made of amino acids. The software helps a person look at proteins, at small molecules, and at documents about drug safety.

A person can upload a PDB file or an mmCIF file. A PDB file is a text file that lists where atoms sit. PDB means the Protein Data Bank, the public archive these files usually come from. mmCIF is a second text format for the same kind of list.

The program counts what is in the file. It draws the molecule in three dimensions. It saves the reading in a local database. It can open that reading again after a restart. It can download a report as a web page, as a text note, or as a data file.

The screen opens on that same computer. The address in the read-me is http://127.0.0.1:8732. The read-me file is README.md.

Each number carries a tag for where it came from. The tags are: taken from the file, reported by a database, calculated, predicted by a model, or typed by the person. An empty measurement stays empty. A warning is stored with it.

Words used here

These words appear in the notes and in the pictures. An angstrom is the length unit used for atoms. One angstrom is a ten-billionth of a metre.

Word	What it means in this note
Protein	A large molecule made of amino acids.
PDB file	A text file of atom positions from the Protein Data Bank. mmCIF is another text format for the same kind of list.
Chain	One continuous string of amino acids in the file.
Residue	One amino acid in a chain. These files also store each water as a residue.
SMILES	A one-line code for a small molecule. Ethanol is CCO.

Word	What it means in this note
Docking score	A placement score in kilocalories per mole (kcal/mol). It comes from a computer. It is not a measured binding strength.
RMSD	Root-mean-square deviation. The typical distance between a predicted pose and the known pose. The unit is the angstrom. A smaller number means a closer pose.
Solvent-accessible surface	The part of the protein a water-sized probe can touch. The unit is square angstroms.
Embedding	A list of numbers that stands in for a sequence, so two sequences can be compared.
Workflow	A saved list of steps that run in order.

Three limits sit behind every result. A docking score is not a measured binding strength. A model result is not a laboratory measurement. A short simulated motion does not show that a protein is stable, that a molecule binds, or that a treatment works.

The milestones

A milestone is one stage of the work. The pictures later in this note use the same numbers. Automated checks are the calculation tests. The source notes call them pytest. Screen checks are tests of the page. The notes call them Vitest. The browser check is a scripted visit to the page. The notes call it Playwright.

Milestone 1. Read a protein file

What a person can do

A person can upload a PDB file or an mmCIF file. The program counts chains, residues, and waters from that file. It draws the molecule. It saves the reading and can open it again after a restart. It can download a report as a web page, a text note, or a data file. Screens for work that was not built yet were marked planned, and they had no actions.

Result

File 1UBQ is titled as a structure of ubiquitin. Chain A has 76 residues in both file formats. The file has 58 water residues, 660 atoms, and 0 hydrogen atoms. The R-free fit statistic is blank in the file, so the program left it blank. File 4HHB is titled as human deoxyhaemoglobin. Its chains have 141, 146, 141, and 146 residues. That is 574 protein residues. Waters are counted separately: 221. A heme group is on chains A, B, C, and D. A phosphate is on chains B and D. Both files record an X-ray experiment. Figure 1 shows the counts. Automated checks: 34 passed and 0 failed. The note does not say how many checks were left out. Screen checks: 7 passed and 0 failed. One browser check passed, and none failed. A health check succeeded. It named the database version 0001_initial. After a restart, the saved ubiquitin reading matched.

Limit

Fold labels, surface area, and hydrogen bonds were not computed in this step. Docking, motion, small molecules, and models were not built yet. The radius of gyration, a single length for how spread out the atoms are, uses the central atom of each residue and does not weight atoms by mass. A chain break is flagged when two of those central atoms are more than 4.5 angstroms apart. That flag is not a crystallographer’s annotation. The setup steps in the read-me are the steps used for this copy. A second copy from an empty folder was not made.

Milestone 2. Measure the protein

What a person can do

A person can paste a protein sequence or fetch one from UniProt, a public sequence archive. The program reports length and an estimated mass. It labels helices and strands. It reports the solvent-accessible surface. It estimates contacts and backbone hydrogen bonds. It can lay one chain on another and report the RMSD. It can search for pockets, which are dents where a small molecule might sit.

Result

The ubiquitin sequence from file 1UBQ is 76 letters long. Its average mass is 8564.736 daltons. A dalton is the unit used for molecular mass. An independent sum of the atoms, minus one water for each join between amino acids, was 8564.737 daltons. The difference is 0.00086 daltons. The solvent-accessible surface of file 1UBQ, with waters left out, was 4893.04 square angstroms by one method and 4834.72 by another. The probe radius was 1.4 angstroms. The two answers differ by 1.19 percent. The limit written down beforehand was 20 percent. Laying chain A on itself gave an RMSD within 0.000001 angstroms, using 76 pairs. A sequence with an illegal character, ACDE1, was rejected, and nothing was stored. On five files kept aside, the pocket center was within 4.0 angstroms of the bound molecule in 3 of 5 cases. Figure 2 shows those five. The same 3 of 5 result was recorded on the five files used to set the method. Automated checks: 55 passed, 0 failed, and 1 left out because it needs a network. Screen checks: 7 passed and 0 failed.

Limit

The pocket finder is a grid estimate. Three successes out of five is the recorded claim. It is not a general test of binding sites. The failures are files 1FKF, 2PK4, 1TLP, and 4DFR. A second pocket program, fpocket, was not installed, so it took no part. Relative surface, as a share of a fully exposed residue, is not reported. No table of maximum exposure is cited. Hydrogen bonds are estimates from backbone atoms. File 1UBQ has no hydrogens. The mass is an average mass with the end water included. It is not a mass from a laboratory instrument. The browser check was not run again. One public record, P62988, is inactive. The program rejects it and reports the successor records. It does not quietly replace the sequence.

Milestone 3. Read a small molecule

What a person can do

A person can enter a small molecule as SMILES. The program checks the formula and the molecular weight. Records brought in from a chemistry database stay separate from numbers the program calculates.

Result

Three checks matched the expected formula and weight. The chemistry toolkit was RDKit 2025.09.1, and the weight was rounded to two decimals. Automated checks: 66 passed, 0 failed, and 1 left out. Screen checks: 7 passed and 0 failed.

Molecule	SMILES	Formula	Weight
Aspirin	`CC(=O)Oc1ccccc1C(=O)O`	`C9H8O4`	`180.16`
Caffeine	`CN1C=NC2=C1C(=O)N(C(=O)N2C)C`	`C8H10N4O2`	`194.19`
Ethanol	`CCO`	`C2H6O`	`46.07`

Limit

Activity rows brought in from ChEMBL, a public database of measured activity, stop at 100 rows. The molecule data file had no licence field, so none was written in. Crippen logP, an estimate of how a molecule splits between oil and water, is an estimate. It is not a label that a molecule is drug-like. The browser check was not run again.

Milestone 4. Place a molecule and score the fit

What a person can do

A person can ask the program to place a small molecule against a protein and read a docking score. The placement program is AutoDock Vina. If Vina is not installed, the program refuses the request. It does not invent a pose.

Result

Vina 1.2.5 ran on file 3PTB with benzamidine, molecule code BEN. The search effort was 4. The seed was 1. A seed is a starting number that lets a run be repeated. The search box was 20 angstroms on a side, centered on the known molecule. The saved output lists three docking scores: −6.073, −5.062, and −4.839 kcal/mol. A second run with the same seed differed by 0.006 kcal/mol. The window written down for that spread was 0.05 kcal/mol. Automated checks: 71 passed, 0 failed, and 1 left out. The note does not say why that one check was left out.

Limit

A screen-check count for this milestone was not recorded. The ten-file pose-distance test was not run in this milestone. It was run later. Figure 3 shows it. A comparison of docking score with measured activity was not run. A wider repeat test, beyond this one pair of runs, was not run. The preparation tool version 0.6.1 does not run on Python 3.12, the language version in use. Version 0.8.0 was installed instead.

Milestone 5. Gather safety documents

What a person can do

A person can review label text, spontaneous report counts, and papers. Each item is tagged by what kind of evidence it is. Text copied from a regulator’s label can be tagged established. A spontaneous report count is observational. It is not proof that a drug caused the event. A look-alike target is a computer prediction and is marked unverified. A paper stays unclassified until a person sets a category. The program does not assign the hypothesis tag by itself.

Result

Sample replies from the openFDA label service, from FAERS spontaneous-report counts, and from PubMed paper lookups were saved on 13 October 2026 and reused as test copies. FAERS is a public list of adverse-event reports. PubMed is a public index of papers. Automated checks: 78 passed, 0 failed, and 1 left out because it needs a network.

Limit

This part has no accuracy score. It does not give a diagnosis or a dose. No workflow for changing a medication was added. At the end of this milestone, Open Targets, a DailyMed backup, Europe PMC, and a full view of alternatives were not built. Those sources were added later as saved copies. A screen-check count was not recorded.

Milestone 6A. List the models

What a person can do

A person can see a registry of prediction models. Each row can name the model, the version, the task, the source, the licence, what the computer must have, and the known limits. A live memory report is included. A model loads only when a person asks. Nothing loads when the program starts.

Result

Automated checks: 88 passed, 0 failed, and 1 left out because it needs a network. Screen checks: 9 passed and 0 failed.

Limit

A list-only loader does not run a model file. This step does not train a model. If a needed program is missing, loading is refused. If the computer does not meet a stated memory or device need, loading is refused.

Milestone 6B. Turn a sequence into numbers

What a person can do

A person can ask for an embedding of a protein sequence. The model is a published model called ESM-2, in a small size marked 8M. The recorded licence is MIT. The registered name is facebook/esm2_t6_8M_UR50D. The weight file is downloaded only when asked, and it is checked against a recorded fingerprint.

Result

Eight local chains were kept, under a rule written before the score was computed: 1UBQ A (76), 4HHB A (141), 4HHB B (146), 3PTB A (223), 1STP A (159), 2GBP A (309), 1HSG A (99), and 1HSG B (99). That is 28 pairs. Rank agreement with sequence similarity was 0.6990131578947367. The low end of the interval is 0.3788637829986782. The high end is 0.8820213682475236. The interval comes from 1,000 resamples, seed 20261013. All 1,000 were finite. Figure 4 shows this one result. Automated checks: 94 passed, 0 failed, and 2 left out (network and model). The optional model check passed 1. Screen checks: 9 passed and 0 failed. The weight file fingerprint matched the published file. The recorded sha256 is 24c5fa474c48f3b754b86efe752d5f189d2bcd88190fa2270fc92b2ef3034189. The file is 31,384,292 bytes.

Limit

This is rank agreement with sequence similarity on that small set. It is not evidence of function. It is not evidence of three-dimensional shape. The 1,000 separate resample values were not saved. Sequences longer than 1,026 positions, counting special markers, are refused. They are not split into pieces. The letter J is allowed in a sequence here, but this model has no token for it, so those sequences are not embedded.

Milestone 6C. Estimate how well a molecule dissolves

What a person can do

A person can train simple estimates, and one small graph network, for log solubility. Log solubility is the logarithm of how many moles dissolve per litre. A mole is a counting unit for molecules. Whole chemical scaffolds stay on one side of the split, so a near-copy of a training molecule is less likely to land in the test. A scaffold is the core ring framework of a molecule.

Result

The public Delaney table was retrieved on 13 October 2026. It has 1,128 data rows and 96,699 bytes. Its sha256 fingerprint is 8c06a76f0c6487d29ab0f903e6a7a7139f189ab3c1178f159c8be8964602f189. The file has no licence line. The software project that hosts the loader uses an MIT licence for the software. That line is not a licence for the table. The split, chosen before fitting, used seed 20261013. Training rows: 902. Test rows: 226. Rows dropped: 0. On the test rows, the typical miss was 1.3217993772506413 for a straight-line fit on simple features, 1.6696791869919676 for a straight-line fit on a chemical fingerprint, and 2.5347372633569814 for a two-layer graph network. The graph network's share of the spread explained is −0.4764081172372039. Zero would mean it did no better than guessing the average of the test labels. Below zero means worse than that average. The run recorded that result. It was not tuned away. Figure 5 shows the comparison. A second run with the same seed matched the typical miss within 0.000001. Automated checks: 98 passed, 0 failed, and 3 left out. The held-out metrics check passed 1. Screen checks: 9 passed and 0 failed.

Limit

These numbers are scores on a published table. They are not solubilities measured for this summary in a laboratory. The screen says so. A later check of whether a new molecule is close to the training set uses a fingerprint similarity cutoff of 0.3. That cutoff was chosen in advance. It was not fit to the test labels. The Delaney file is not stored in the project. Training checks the fingerprint above. The note does not say why the three left-out checks were left out.

Milestone 6D. Relax a structure, then take a few steps

What a person can do

A person can prepare a structure, lower its stored energy, and run a short motion. The motion library is OpenMM. Preparation fills missing atoms and removes extra molecules. That changes the file, and the notes say so. Relaxation lowers stored energy. It does not move the protein through time.

Result

OpenMM 8.6.1 on this machine offered two ways to calculate: a reference mode and the main processor. No graphics-card mode was listed. A five-step relaxation of file 1UBQ lowered the stored energy. The start and end energies of that five-step run were not recorded. The check only says the end is lower than the start. A separate 20-step relaxation the same day went from −328.93633723496714 kJ/mol to −8331.638644503953 kJ/mol. A kilojoule per mole is an energy unit. One constant-energy run of 20 steps had a relative energy drift of 0.0004987123454089311. An earlier run the same day had a drift of 0.08066315438678218. The planned limit was 0.02. It is not treated as met, because the two runs do not both sit under it. Figure 6 shows these endpoints. A different short run, with a thermostat, had a mean temperature of 260.1331467073965 kelvin. The planned window was 150 to 450 kelvin. Two runs with the same seed differed by 1.6878309150943096 kJ/mol in final total energy. The planned match was within 1.0 kJ/mol. That match was missed. Automated checks: 103 passed, 0 failed, and 3 left out. Screen checks: 9 passed and 0 failed.

Limit

At the end of this milestone, restarting from a saved checkpoint was not built. A person can run at most 500 motion steps, and at most 100 relaxation steps. There is no graphics processor on this machine. A short run does not show stability, binding, or a treatment effect. The note does not say why the three left-out checks were left out.

Milestone 6E. Recombine molecules you supply

What a person can do

A person can recombine pieces of molecules they already supplied. The method is called BRICS. Each drawing is labeled a computational hypothesis. Three liability filters, named PAINS, Brenk, and NIH, stay on. A request that mentions toxicity or hazard is rejected.

Result

Automated checks: 105 passed, 0 failed, and 3 left out. Validity, uniqueness, novelty, diversity, and the filter pass rate are calculated for each request, together with the seed.

Limit

This is not a trained model that invents molecules from scratch. The filters cannot be turned off. The drawings are not synthesized and not tested in a laboratory. No fixed benchmark number for those rates was written down. A screen-check count was not recorded. The browser check was not run again.

Milestone 6F. Run a saved list of steps

What a person can do

A person can save a workflow. The steps the program already knows are: read a structure, describe a small molecule, or dock. If one step succeeds and a later step fails, a resume can skip the success and rerun the failure. If the docking program is missing, docking is skipped, and so are steps that depend on it. A written summary from a language model stays off unless a switch named BEAI_LLM_ENABLED is turned on.

Result

A run on a hand-built peptide plus ethanol returned a real residue count and the canonical SMILES CCO for ethanol. Canonical means the standard form of that one-line code. With the docking program hidden, docking and the dependent step were skipped. With the summary switch off, the summary request was refused. With the switch on, a paper id that was not in the retrieved set was rejected. The test used PMID:9 when only PMID:1 had been retrieved. PMID is a public number for a paper. Automated checks: 109 passed, 0 failed, and 3 left out (network and model). The screen’s production build completed.

Limit

A workflow does not draw a scientific conclusion. Each step keeps its own tag for where the number came from. The peptide residue count from that run was not written down. A screen-check count was not recorded. The browser check was not run again. The recorded real docking step inside a workflow is in the closing check below.

Closing check. Fill the gaps that were still open

What a person can do

A person can see a pose distance for every file in a set of ten, including the failures. The person can read saved copies of three more safety sources: DailyMed, Europe PMC, and Open Targets. The person can restart a short motion from a saved point and compare it with a run that was not stopped. A workflow can start Vina when Vina is installed.

Result

The ten-file redock used Vina 1.2.5 on 13 October 2026. Search effort was 8. The seed was 1. One pose was kept. The box was 20 angstroms on a side. On the files used to set the pocket method, 3 of 5 poses were within 2.0 angstroms. On the files kept aside, 2 of 5 were within 2.0 angstroms. Files 2PK4, 4DFR, and 1DWD failed preparation and have no RMSD. Figure 3 shows every file. A focused safety check passed 12, with 0 failed. The checkpoint check passed 1. The largest coordinate difference versus a continuous run was 7.993713069254227e-08 angstroms. The planned bound was 0.00001 angstroms. The recorded energy drifts were not changed. Workflow checks plus one docking check passed 6, with 0 failed. That run included a real Vina step and the hidden-program skip. The full automated suite was 118 passed, 0 failed, and 3 left out (network and model). Screen checks: 9 passed and 0 failed. The screen build completed.

Limit

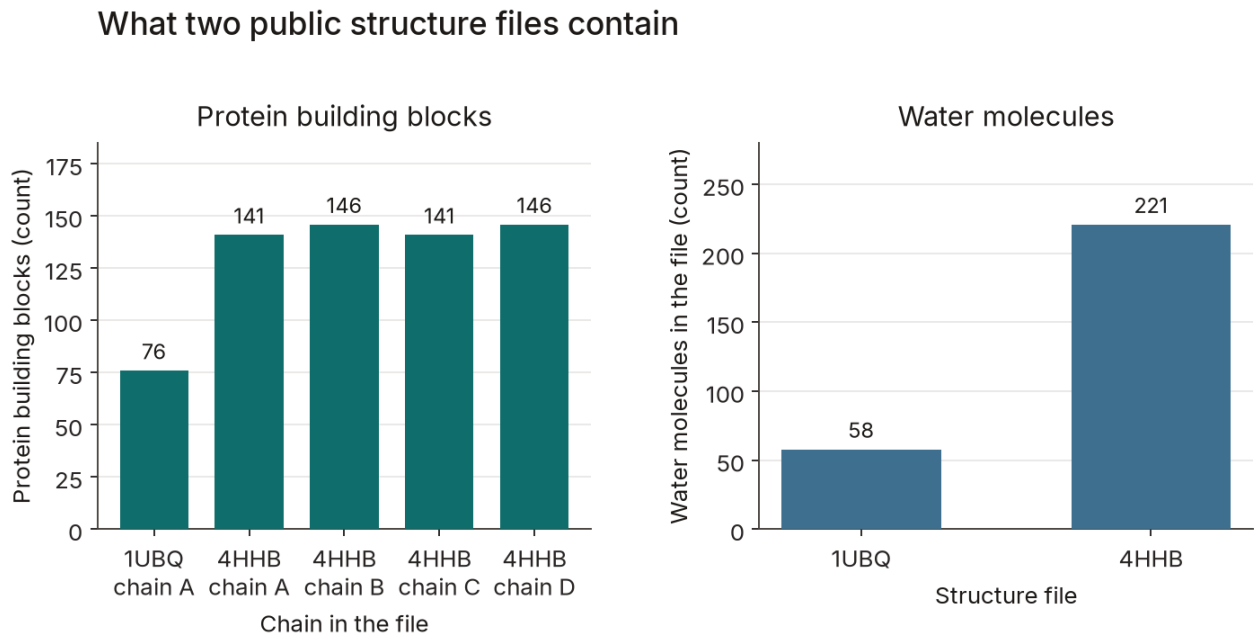
fpocket is still absent. A search of the package list found no fpocket package. The energy-drift limit of 0.02 was left unmet. The browser check was not run again. DailyMed is the backup when an openFDA label fetch fails. The warfarin search and label saved on 13 October 2026 copy section text from that label. For Europe PMC, the aspirin record is not open access, so its abstract is not stored. An open-access record keeps the abstract and the licence line “cc by”. Open Targets safety rows and disease rows for VKORC1 stay unclassified. A failed request stores no row.

Pictures of the recorded numbers

There are seven pictures. Bar heights use the recorded values. A missing bar was not computed. It is not zero. Where a sentence earlier in this note shortens a long decimal, the table prints every digit that was written down.

Figure 1. What two structure files contain

The left picture counts protein residues, the amino-acid units in each chain. Water, heme, and phosphate are not in those bars. The right picture counts water residues. In these two files that count matches the public solvent-atom count. Both axes start at zero.



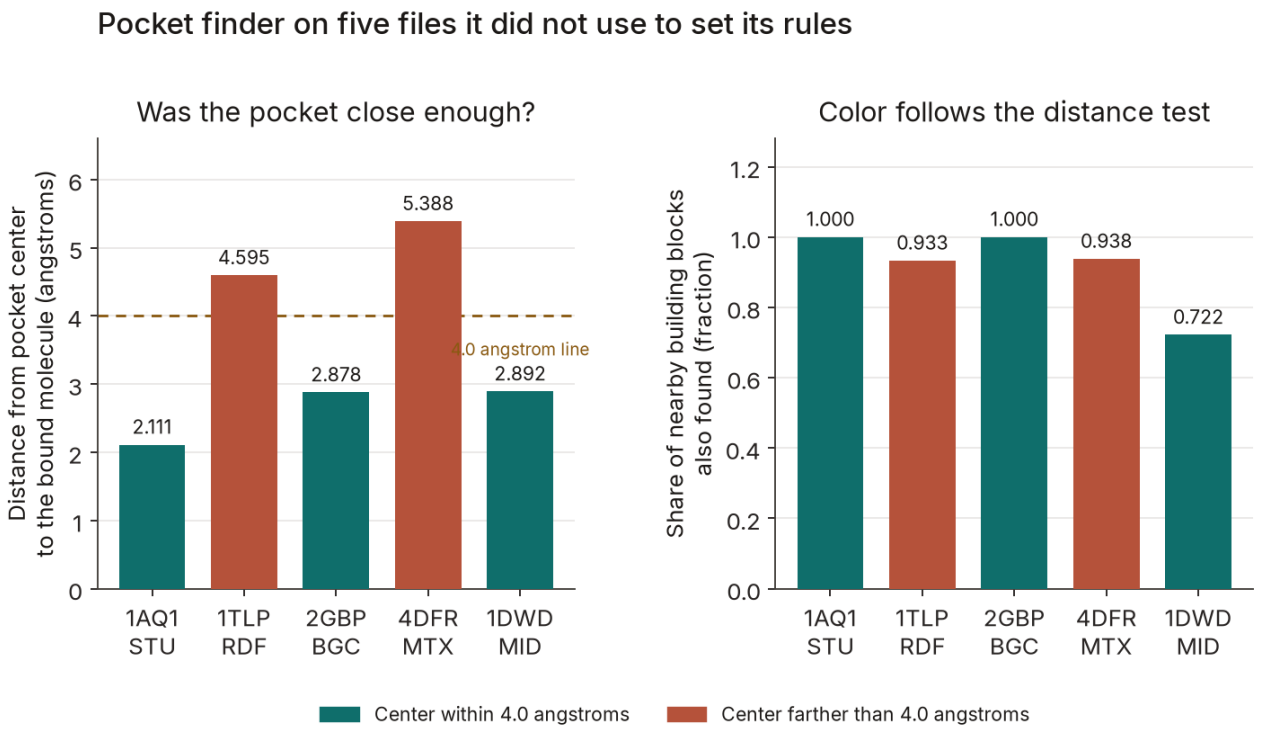
Both axes start at zero. Left: polymer residues (amino-acid units). Water, heme, and phosphate are not in those bars. Right: water residue counts. They match the public solvent-atom counts. Source: scientific validation document, structure overview, passed 11 October 2026. The same counts are in the Milestone 1 status note.

Figure 1. Counts from the scientific validation document, structure overview, passed 11 October 2026. The same integers are in the Milestone 1 status note and in the public-record snapshot copied that day.

Figure 2. Pocket finder on files kept aside

Success was written down before the test. One of the top three pockets must have its center within 4.0 angstroms of the center of the bound molecule. The bound molecule is the largest extra residue that is not a buffer chemical. The right picture is the share of protein residues near that molecule that also line the closest of those pockets. Color follows the distance test. A tall red bar can still be a miss.

The grid step is 1.0 angstrom. Protein atoms are treated as spheres of radius 2.5 angstroms. The search looks in a shell 5.0 angstroms deep. A point must be empty in at least 5 of 7 directions. A pocket needs at least 8 grid cubes. Lining residues lie within 5.0 angstroms. An earlier radius of 1.7 angstroms and a rule of 4 directions left the protein core unfilled on the tuning files, so those two settings were changed using only those files. The files kept aside were not used for that change. The source notes call the first group tuning and the second group held-out.



Bar labels are the values printed in the source table. A red bar on the right can still be high: color follows distance, not recovery.
Source: scientific validation document, pocket table, rows kept aside, run 13 October 2026. The pass mark was written down first.

Figure 2. Held-out rows from the pocket table in the scientific validation document. Run on 13 October 2026. Bar labels are the printed table values.

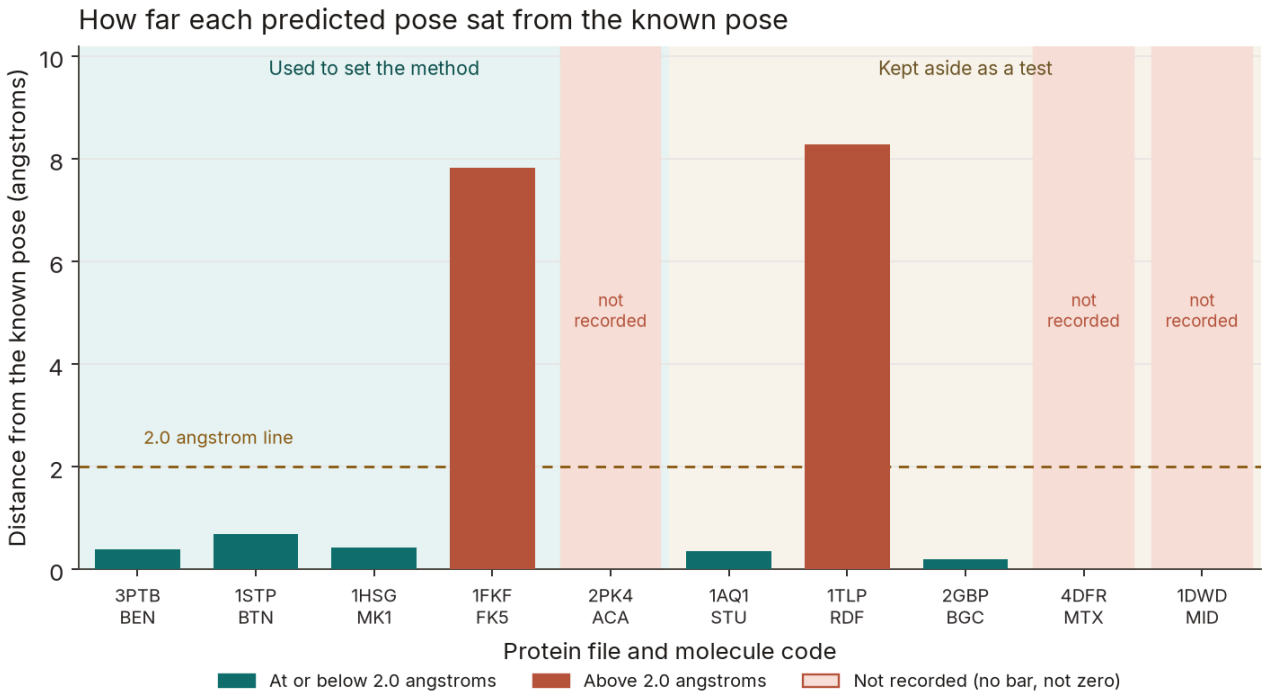
All ten pocket results. Shaded rows missed the 4.0 angstrom line.

Group	File	Code	Distance (angstroms)	Within 4.0	Share found
Tuning	3PTB	BEN	0.565	yes	1.000
Tuning	1STP	BTN	2.245	yes	0.750
Tuning	1HSG	MK1	0.675	yes	1.000
Tuning	1FKF	FK5	5.460	no	0.643
Tuning	2PK4	ACA	12.835	no	0.000
Kept aside	1AQ1	STU	2.111	yes	1.000
Kept aside	1TLP	RDF	4.595	no	0.933
Kept aside	2GBP	BGC	2.878	yes	1.000
Kept aside	4DFR	MTX	5.388	no	0.938
Kept aside	1DWD	MID	2.892	yes	0.722

Tuning success was 3 of 5. Held-out success was 3 of 5. Source: scientific validation document, pocket table.

Figure 3. Distance from the known pose

This is the ten-file test from the closing check, not the single benzamidine score check in Milestone 4. RMSD here compares heavy atoms, which means atoms other than hydrogen, in the original frame. Symmetry is accounted for, so a spun but identical ring is not counted as a miss. The pose was not slid onto the known molecule before the measurement. The pass line is 2.0 angstroms.



Bar heights are the full recorded distances. A missing bar was not computed. It is not zero. The three missing poses failed preparation.
Source: scientific validation document, redocking table, 13 October 2026. The closing status note repeats 3 of 5 and 2 of 5, and the three gaps.

Figure 3. Redocking table in the scientific validation document, 13 October 2026. The closing status note repeats the 3 of 5 and 2 of 5 counts, and the three files with no distance.

File 2PK4 stopped during preparation. The recorded reason is an extra bond between residues at chain A, position 32. File 4DFR stopped the same way at chain B, position 131. File 1DWD stopped during preparation. The recorded note says CGLN, with 9 heavy atoms missing, none extra, and no missing or extra bonds. On files that did produce a pose, a bond-order step warned that more than one pattern matched and it picked one. The docking score column is a placement score. It is not a measured binding strength.

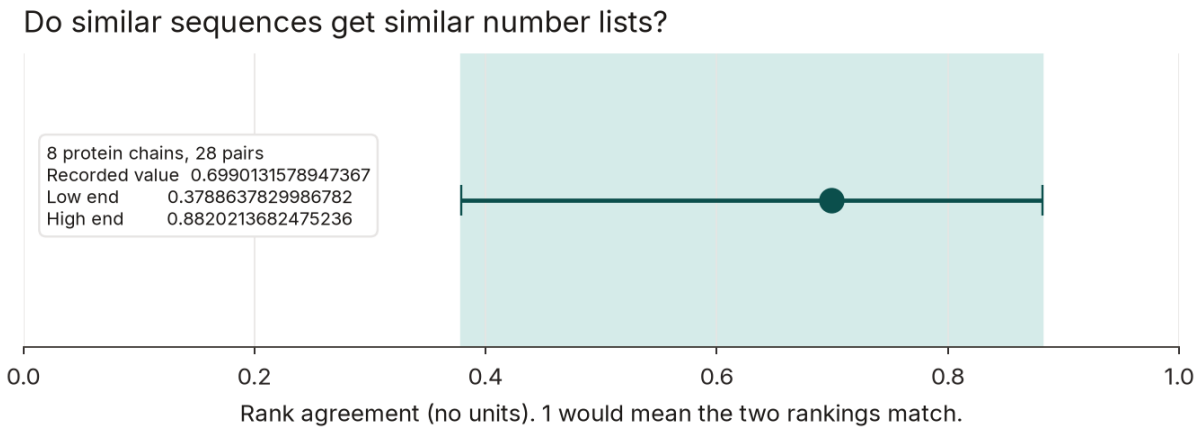
Every recorded digit. Shaded rows missed 2.0 angstroms or have no distance.

Group	File	Code	Distance (angstroms)	Within 2.0	Score (kcal/mol)
Tuning	3PTB	BEN	0.38634036044964726	yes	-6.196
Tuning	1STP	BTN	0.6946867472014318	yes	-7.04
Tuning	1HSG	MK1	0.42136429418226423	yes	-11.567
Tuning	1FKF	FK5	7.823345721495985	no	-6.548
Tuning	2PK4	ACA	not computed	no	not computed
Kept aside	1AQ1	STU	0.3478932885966152	yes	-13.841
Kept aside	1TLP	RDF	8.278915104289972	no	-8.39
Kept aside	2GBP	BGC	0.19162198907988706	yes	-7.357
Kept aside	4DFR	MTX	not computed	no	not computed
Kept aside	1DWD	MID	not computed	no	not computed

Source: scientific validation document, redocking table. Scores are copied as printed.

Figure 4. Sequence number lists

The dot is one rank-agreement score. Rank agreement asks whether the order of the 28 pairs by embedding similarity matches their order by sequence similarity. The shaded band is the recorded interval. There is no plot of the 28 pairs, because those pair values were not written down.



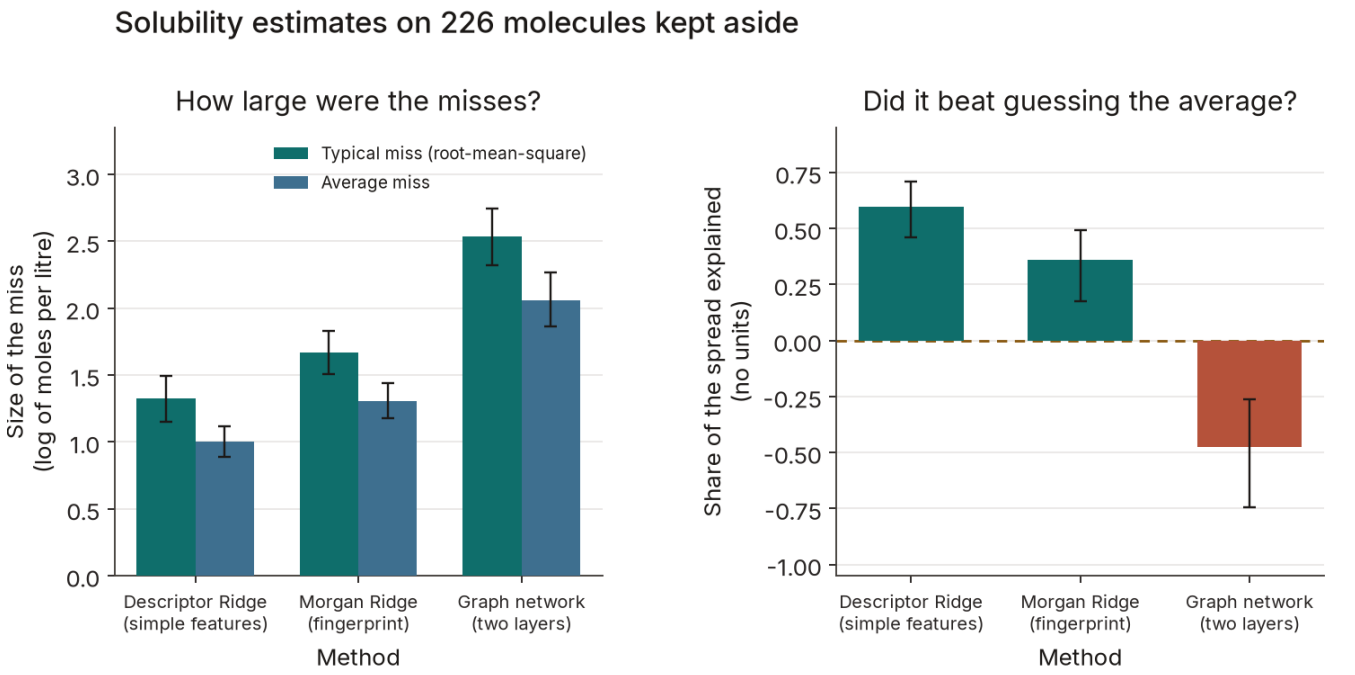
The dot is the recorded rank agreement. The line and the shaded band are the 2.5 and 97.5 percentiles of 1,000 resamples. Seed 20261013. The 1,000 separate values were not saved. All 1,000 were finite. This is not a claim about function or shape. Source: scientific validation document, embeddings section. The same three numbers are in the remaining plan status note.

Figure 4. Embeddings section of the scientific validation document. The same three numbers are in the remaining plan status note. Seed 20261013. All 1,000 resamples were finite.

Item	Recorded value
Rank agreement	`0.6990131578947367`
Low end (2.5 percentile)	`0.3788637829986782`
High end (97.5 percentile)	`0.8820213682475236`
Chains, and pairs	`8 chains, 28 pairs`
Resamples	`1,000, seed 20261013, all finite`

Figure 5. Solubility estimates

The left picture is the size of the miss on 226 test molecules. The typical miss squares each error, averages the squares, and takes the square root, so large misses count extra. The average miss does not square them. The right picture asks how much of the spread the method explained. Zero means the same as guessing the average test value every time. Whiskers are the 2.5 and 97.5 percentiles of 1,000 resamples. They are not a laboratory error bar.



Whiskers are the recorded 2.5 and 97.5 percentiles of 1,000 resamples (seed 20261013), not a lab error bar.
Bar heights are the full recorded values. Every digit is in the summary table. The graph network is below zero on the right.
Source: scientific validation document, solubility table. The same point estimates are in the remaining plan status note.

Figure 5. Solubility table in the scientific validation document. The same point estimates are in the remaining plan status note. The graph network sits below zero.

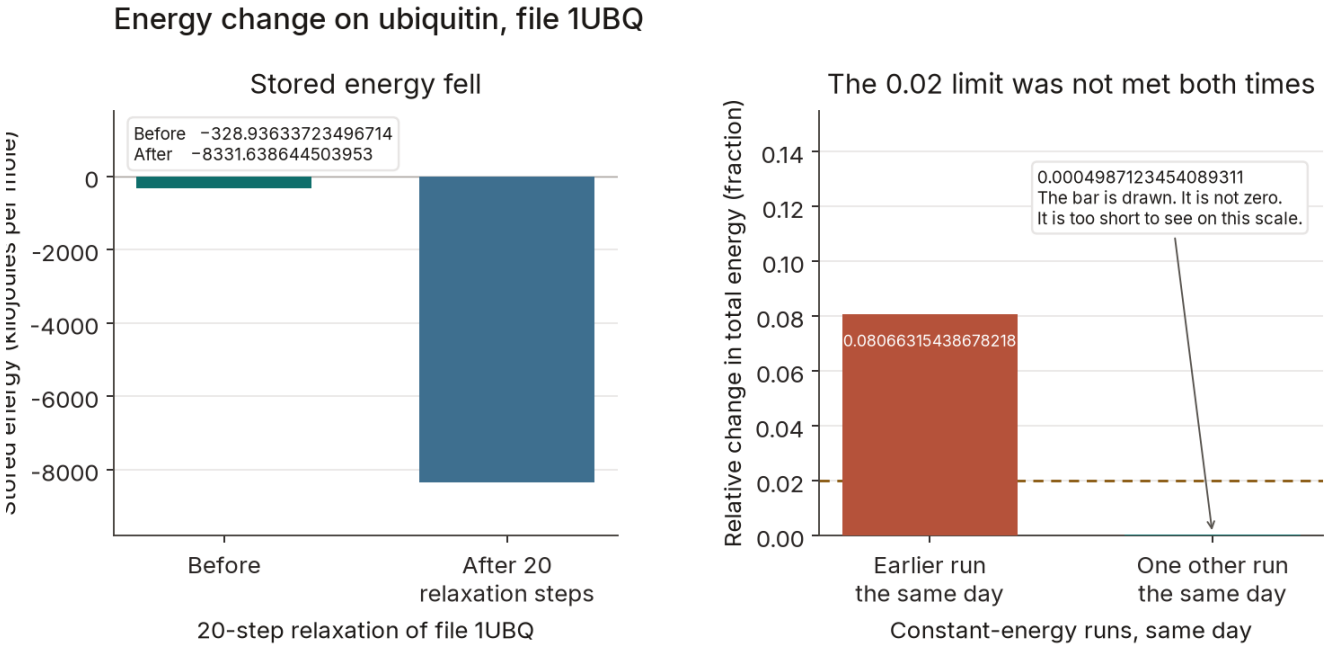
Every recorded digit. The graph-network rows are shaded.

Method	Score	Recorded value	Low end	High end
Descriptor Ridge	Typical miss	1.3217993772506413	1.147560194604866	1.4920541722551173
Descriptor Ridge	Average miss	1.0004937427916745	0.8907326010644745	1.1141110380395272
Descriptor Ridge	Spread explained	0.5985128816471968	0.45973007459714554	0.7074329222377606
Morgan Ridge	Typical miss	1.6696791869919676	1.506008911536875	1.8300778122281194
Morgan Ridge	Average miss	1.3029135097542737	1.1760270767204246	1.4366200664112465
Morgan Ridge	Spread explained	0.3593709962979724	0.1757942096236605	0.4941057951191802
Graph network	Typical miss	2.5347372633569814	2.3179658213153536	2.7474698037745178
Graph network	Average miss	2.056036578614957	1.8600800380376539	2.264292208255532
Graph network	Spread explained	-0.4764081172372039	-0.7456244760034777	-0.2633770633818176

The straight-line fits are Ridge fits. Their penalty strength was 1.0 and was not tuned. The fingerprint is a Morgan fingerprint, radius 2, with 2,048 yes-or-no features. The graph network has two layers, a hidden size of 32, 20 passes through the training rows, and a learning rate of 0.01. None of those choices were tuned on the test rows. The older predicted column in the Delaney file was not used. The loader names the table as Delaney, Journal of Chemical Information and Computer Sciences, volume 44 (2004), pages 1000–1005.

Figure 6. Energy change on file 1UBQ

The left picture is the 20-step relaxation. Only the two endpoints were saved, so there is no step-by-step line. The right picture is the relative change in total energy for two constant-energy runs. That motion holds the particle count, the volume, and the energy steady. The source notes call it NVE. It ran 20 steps of 1 femtosecond, which is one millionth of a nanosecond, with zero starting speeds, after 20 relaxation steps. The dashed line is the planned limit of 0.02. One run is under it. An earlier run the same day is over it.



Only these endpoints were saved. There is no step-by-step trace. The dashed line is the planned limit of 0.02.
One run is under it and an earlier run is over it, so the limit is not treated as met. Labels are the full recorded values.
Source: scientific validation document, motion section. The closing status note says the drift numbers were left as recorded.

Figure 6. Motion section of the scientific validation document. The closing status note says these drift numbers were left as recorded.

Shaded row: the limit that is not treated as met.

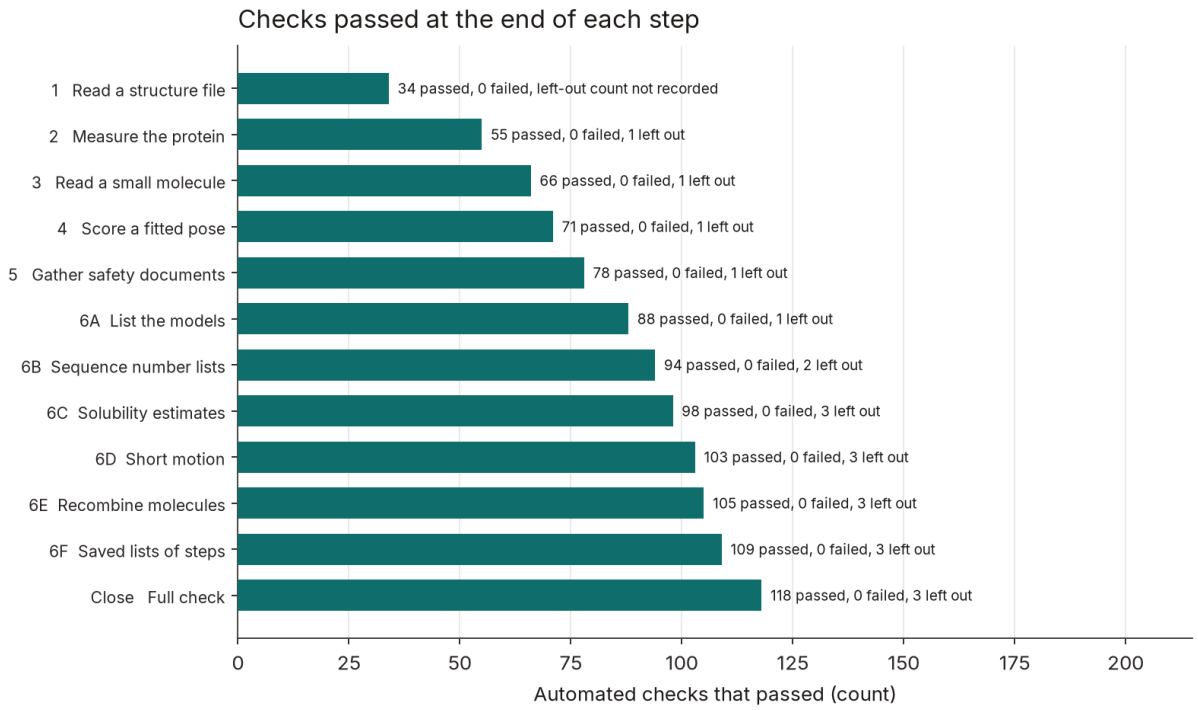
What was recorded	Value
20-step energy, before	`-328.93633723496714 kJ/mol`
20-step energy, after	`-8331.638644503953 kJ/mol`
5-step energies	`not recorded; the check says the end is lower`
Earlier drift	`0.08066315438678218`
Earlier total energy, start	`-694.7754278681138 kJ/mol`
Earlier total energy, end	`-750.818205470382 kJ/mol`
Other run, drift	`0.0004987123454089311`
Other run, start	`-10170.949423790193 kJ/mol`
Other run, end	`-10176.021801832367 kJ/mol`
Planned drift limit	`0.02, not treated as met`
Mean temperature	`260.1331467073965 kelvin`
Temperature window	`150 to 450 kelvin, and this run is inside it`

What was recorded	Value
Same-seed energy gap	1.6878309150943096 kJ/mol
Planned energy match	1.0 kJ/mol, missed
Checkpoint coordinate gap	7.993713069254227e-08 angstroms
Checkpoint bound	0.00001 angstroms, and the gap is inside it

Preparation used PDBFixer 1.12.0 at pH 7.0. The force-field files were amber14-all.xml and implicit/gbn2.xml. Forces between distant atoms used no cutoff. Bonds to hydrogen were held fixed. The run used one processor thread. The temperature run was 15 steps of 1 femtosecond, seed 7, after 15 relaxation steps. The checkpoint check reloaded one prepared file, because preparing the file twice does not give the same coordinates. It used seed 11, 2 relaxation steps, and 4 steps of 1 femtosecond. It did not change the drift numbers. Source: scientific validation document, motion section.

Figure 7. Checks passed at the end of each step

Each bar is the full run at that point. It is not a score, and it is not the number of new checks. The count grew because more checks were added. Every one of these runs recorded 0 failed. A geometry run of 31 passed and 0 failed, on 11 October 2026, is recorded in the structure section. That run is not a bar. The first bar uses the finished Milestone 1 count of 34. The note does not say whether 31 is part of the 34.



Each bar is the full run at that point, not the number of new checks. Every run recorded 0 failed. The count grew because the suite grew.
First bar: Milestone 1 status note. A geometry run of 31 passed is not a bar. Other bars: remaining plan status note.
"Left out" means checks skipped on purpose. The first note does not say how many were left out.

Figure 7. First bar: Milestone 1 status note. Other bars: remaining plan status note. Left out means checks skipped on purpose.

Screen checks, where a count was written down.

Step	Screen checks	Browser check
1. Read a structure file	7 passed, 0 failed	1 passed, 0 failed
2. Measure the protein	7 passed, 0 failed	not run again

Step	Screen checks	Browser check
3. Read a small molecule	7 passed, 0 failed	not run again
4. Score a fitted pose	not recorded	not stated in that note
5. Gather safety documents	not recorded	not stated in that note
6A. List the models	9 passed, 0 failed	not stated in that note
6B. Sequence number lists	9 passed, 0 failed	not stated in that note
6C. Solubility estimates	9 passed, 0 failed	not stated in that note
6D. Short motion	9 passed, 0 failed	not stated in that note
6E. Recombine molecules	not recorded	not run again
6F. Saved lists of steps	not recorded	not run again
Closing check	9 passed, 0 failed	not run again

Where a note records a production build of the screen, that build completed. The notes record a completed build for every step in the table above. The browser used for the one passing visit unpacked to about 681 megabytes. The downloaded archives were 191.1 megabytes, 120 megabytes, and 2.2 megabytes. Firefox and WebKit were not installed.

Other checks that were written down

These are single checks. They are not drawn as pictures. All of them are in the scientific validation document.

Check	What was recorded
Spread of two points	Two equal points 3.0 angstroms apart give an unweighted radius of gyration of 1.5 angstroms. The tolerance was 0.000001. This passed in a run of 31 checks, with 0 failed, on 11 October 2026.
Hand-built peptide	Central-atom distances 3.0 and 5.0 angstroms. The 5.0 angstrom pair is a chain break at the 4.5 angstrom threshold. Residue A2 is missing a CB atom. A missing-residue note, REMARK 465 GLY A 3, is reported. Three central atoms at positions 0, 3, and 8 give a radius of gyration of sqrt(98/9).
Fold labels, file 1UBQ	Program mkdssp 4.2.2. Helix H 12, 3-10 helix G 6, pi helix I 0, strand E 24, bridge B 2, turn T 12, bend S 4, polyproline P 4, loop 12. The file's helix notes cover residues 23–34 (class 1) and 56–59 (class 5). Overlap: both helix 15, program-only helix 3, file-only helix 1; both strand 24, program-only strand 2, file-only strand 9. The counts are not expected to match. The program uses hydrogen-bond patterns. The file uses the depositor's notes.
Self fit	An exact copy, a copy turned by 90 degrees, and file 1UBQ chain A against itself were each within 0.000001 angstroms. A two-residue chain was refused. The minimum is 3 pairs.
Illegal sequence	ACDE1 was rejected. The sequence list stayed empty.

Values that were not recorded

These items are named in the source notes. No number was written down for them. This summary does not fill them in.

- Docking score compared with measured activity, for one target and one activity type: not run.
- A wider docking repeat test, beyond the single benzamidine pair: not run.
- Start and end energies of the five-step relaxation: not recorded. The check only says the end is lower.
- Energy at every step of the constant-energy motion: not recorded. The two drifts and the endpoint energies are the record.

• The 1,000 separate rank-agreement resamples: not recorded. The two percentiles, and the statement that all 1,000 were finite, are the record.

• The 28 pair-by-pair similarities: not printed. Only the one rank agreement and its interval were printed.

• Fixed scores for validity, uniqueness, novelty, diversity, and filter pass rate: not recorded. They are computed for each request.

• The residue count from the peptide-and-ethanol workflow: not written down. The ethanol code CCO was written down.

• Screen-check counts for milestones 4, 5, 6E, and 6F: not recorded.

• How many checks were left out of the Milestone 1 run of 34: not recorded.

• Why one check was left out in Milestone 4, and why three were left out in the solubility and motion runs: not stated next to those counts.

• An atom count for file 4HHB: not in the copied public fields used for these checks. It is omitted.

• A second empty copy of the project, to repeat the setup steps: not made.

Limits that still apply

fpocket is still absent. Pocket jobs skip that stage and say so. The grid pocket finder still runs. Held-out success of 3 of 5 is the recorded claim.

The planned energy drift of 0.02 was not stable on file 1UBQ. The same-seed energy gap missed 1.0 kJ/mol. A short run is not evidence of stability, binding, or a treatment effect.

One section of the limits document still says OpenMM, PyTorch, and Vina are not installed, and that the obabel chemistry tool is missing from the command path. Later sections, and the validation document, record the OpenMM, torch, and Vina runs plotted here. This summary follows those later runs. Later notes do not record installing obabel. No graphics processor is recorded. The motion library did not list a graphics-card mode.

The machine note records 15 GiB of memory, about 4.9 GiB free, and no swap space. A GiB is a unit of memory. Embeddings longer than the model limit are refused. A model whose stated need exceeds the free memory must be refused. The stated 512 MiB floor for the small protein model is not a measured peak.

• Relative solvent-accessible surface is not reported.

• Hydrogen bonds are backbone estimates. File 1UBQ has no hydrogens.

• The overview radius of gyration is unweighted over central atoms.

• Chain breaks use a 4.5 angstrom default. They are not crystallographic annotations.

• Sequence mass is an average mass with terminal water included.

• Embeddings are not experimental measurements and are not evidence of function or shape.

• Solubility outputs are predictions. The Delaney file has no licence line and is not stored in the project.

• A docking score is not a measured binding strength. Score versus activity was not run.

• Safety evidence has no accuracy target. The pages are a research review. They are not a dosing tool.

• Generation outputs are hypotheses. The filters cannot be turned off.

• A workflow does not draw a scientific conclusion. The language-model summary stays off unless the switch is set.

• No automatic check on a shared build service is recorded. Long jobs run inside the same program.

• Preparation removes extra molecules and adds atoms. That changes the input structure.

• The read-me setup is the sequence used to create this environment. A second empty copy was not made.

Where the numbers come from

Every number in this note comes from one of these documents. The pictures use the same sources.

README.md
What the running program does, the screen address, and the size of the browser-check download.

docs/SCIENTIFIC_VALIDATION.md
Structure counts, the pocket table, the redocking table, the compound checks, the rank agreement, the solubility table, and the energy, temperature, and checkpoint numbers. Figures 1 through 6 use this file.

docs/ARCHITECTURE.md
How evidence is tagged, and the versions of the chemistry toolkit, the motion library, and the model tools.

docs/KNOWN_LIMITATIONS.md
Limits that still apply, including the missing pocket program, the unmet energy limit, and the browser check that was not run again.

backend/tests/fixtures/rcsb_entry_snapshot.json
Public-record fields copied on 11 October 2026. They match the file 1UBQ and file 4HHB counts in Figure 1, including the titles ubiquitin and human deoxyhaemoglobin. This file is not a source of extra plotted numbers.

Milestone 1 status note
The first milestone’s check counts: 34 passed and 0 failed, 7 screen checks, and 1 browser check. The same structure counts are repeated there.

Remaining plan status note
Check counts for milestones 2 through 6F and the closing run. Figure 7 uses this note for every bar after the first. The rank agreement, the solubility point estimates, and the energy drifts match the validation document.